

QuickCart Warehouse Inventory Stockout Risk

Supervised ML Classification — Final Project Report

1. Executive Summary

This project predicts Safe, At-Risk or Imminent stockout risk for every SKU, store and day. The final workflow combines relational joins, data-quality handling, feature engineering, temporal validation and class-imbalance-aware modeling.

2. Data and Business Problem

The supplied specification defines five tables: 12 stores, 60 SKUs, 15 suppliers, 30 event days and 21,600 store-SKU-day fact rows. The target is imbalanced: Safe 65.42%, At-Risk 24.01%, Imminent 10.57%. Missing an Imminent stockout is operationally more costly than a false alarm.

3. Data Quality

Check	Result
Stores	12
SKUs	60
Suppliers	15
Events	30
Fact rows	21,600
Lead-time actual missing	19,899 (92.1%)
Supplier reliability missing	3 suppliers
City display case issues	ST03 and ST09

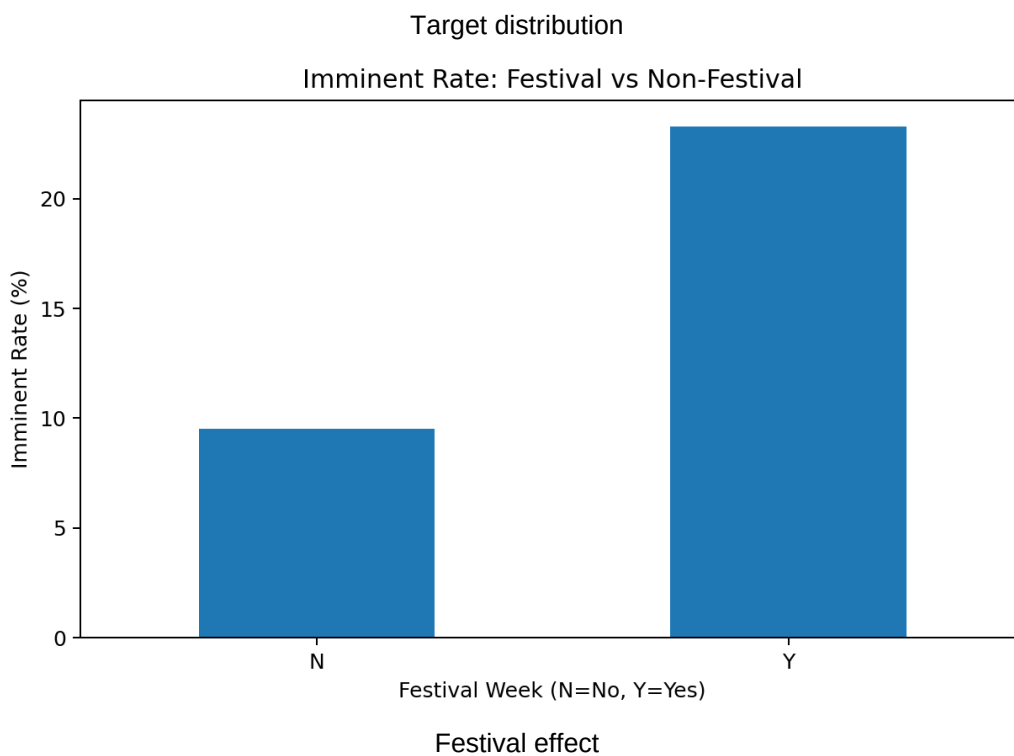
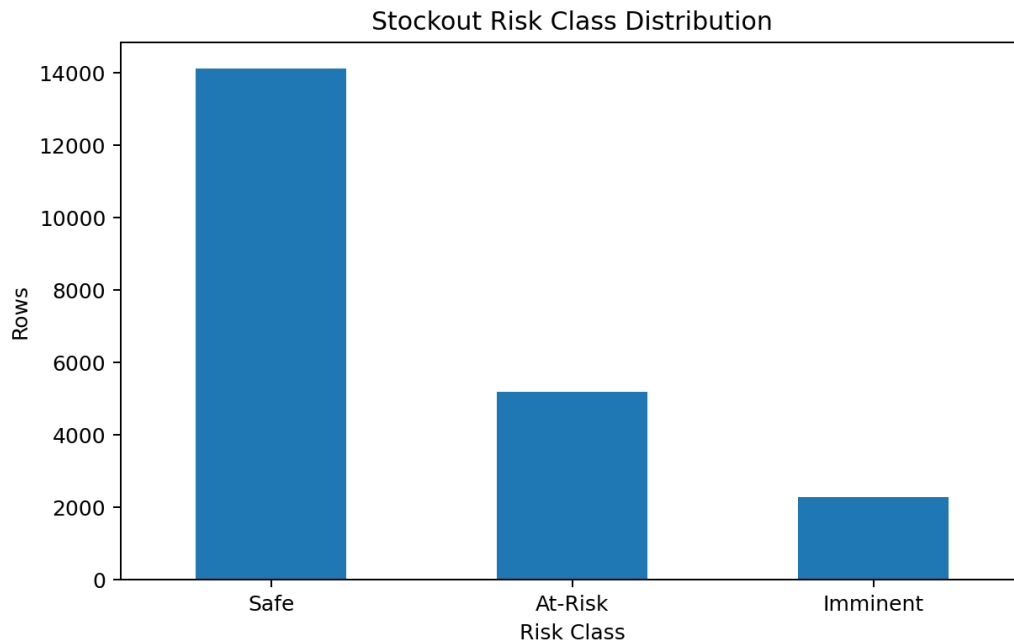
4. Feature Engineering

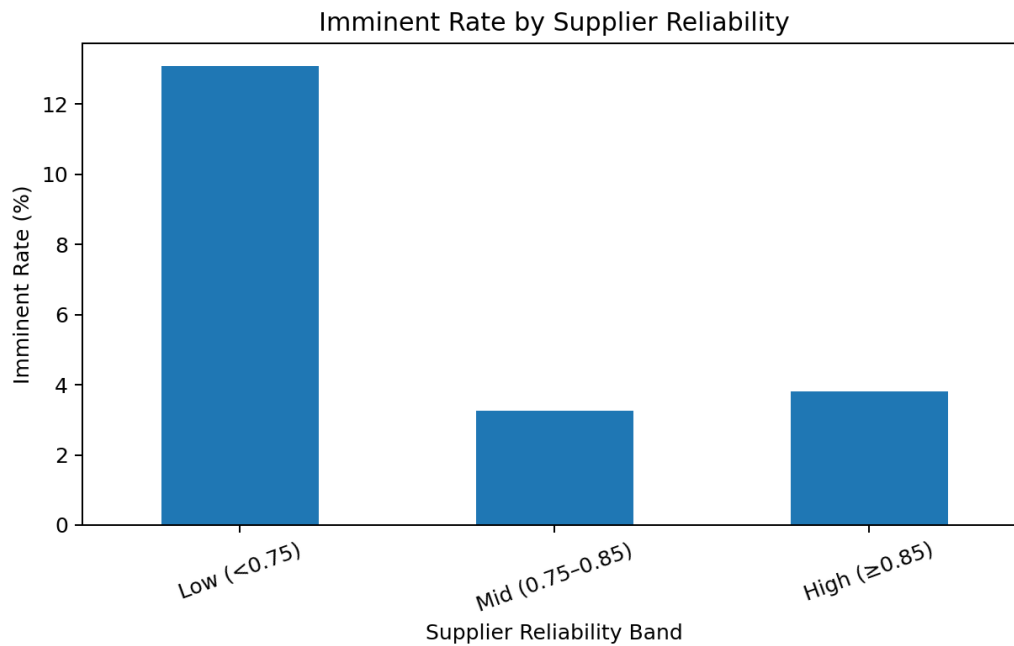
Features include reorder_gap, days_of_cover_ratio, cleaned supplier reliability, prior 3-day reorder activity, day-of-month, days since festival start and an effective event demand multiplier. Categorical fields are one-hot encoded; numeric fields are median-imputed and scaled.

5. Validation Strategy

The final test uses Oct 24–30, while training uses Oct 1–23. An earlier Oct 19–23 validation window is used to choose the Imminent probability threshold, avoiding test-set threshold tuning.

6. Exploratory Findings



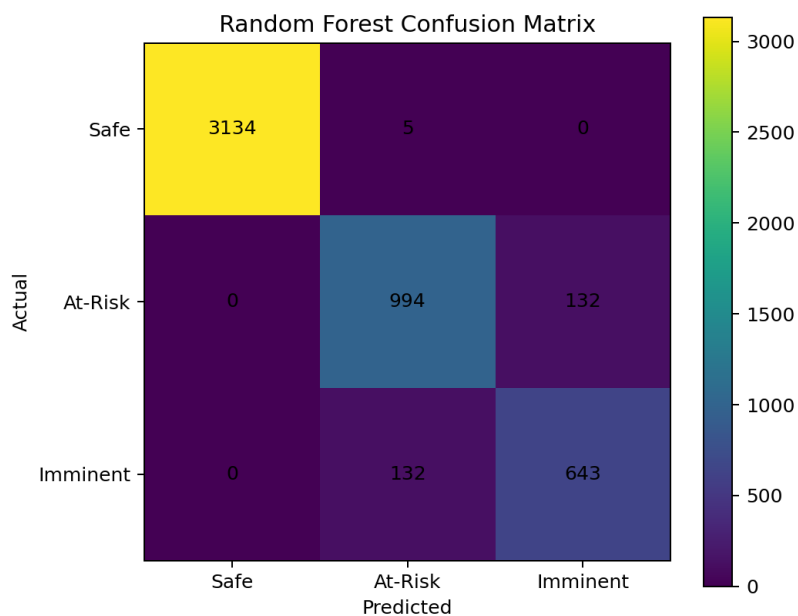


Supplier reliability effect

7. Model Comparison

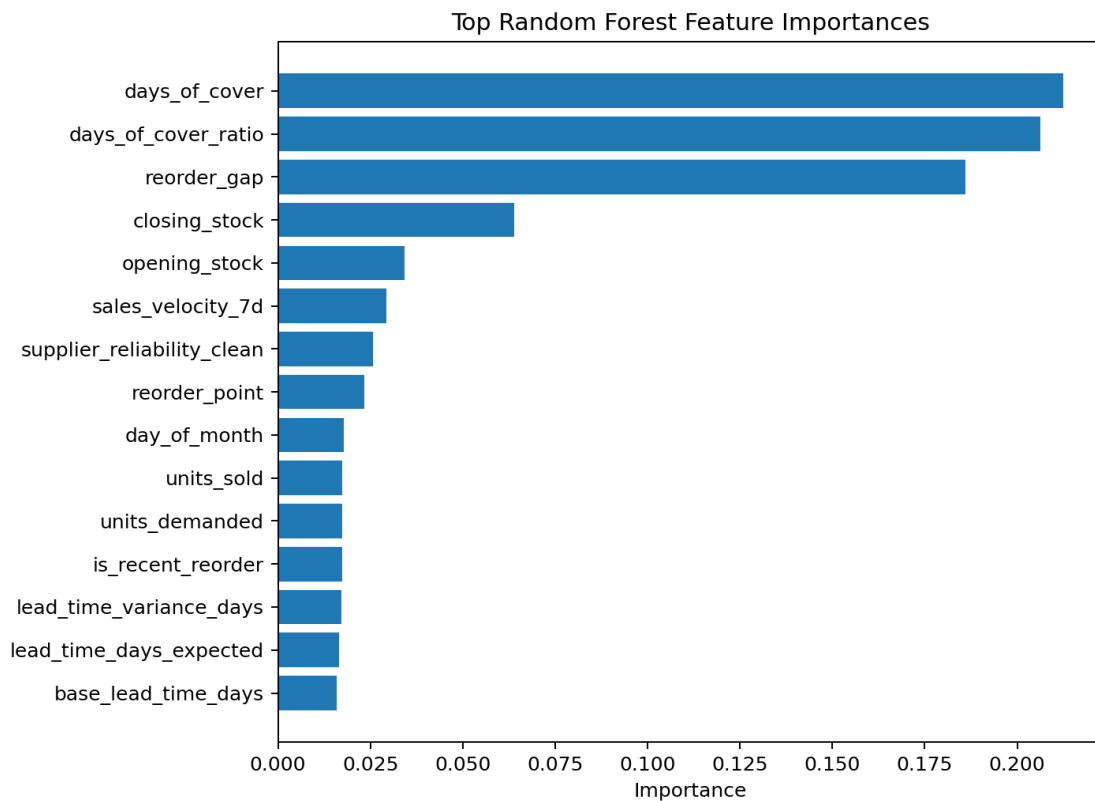
Model	Accuracy	Balanced Acc.	Imminent Recall	Imminent Precision	Macro F1
Majority baseline	62.28%	33.33%	0.00%	0.00%	25.59%
Logistic Regression	82.28%	73.85%	99.61%	48.46%	66.79%
Random Forest + 0.35 threshold	94.66%	90.36%	82.97%	82.97%	90.32%

The selected Random Forest achieved 94.66% accuracy and 82.97% recall for Imminent on the final test period.



Final-test confusion matrix

8. Feature Importance



The leading model signals are `days_of_cover`, `days_of_cover_ratio` and `reorder_gap`, followed by closing stock, opening stock, sales velocity and supplier reliability.

9. Business Recommendations

- Use Imminent predictions as the urgent replenishment queue.
- Review At-Risk items before their expected delivery window.
- Increase safety-stock attention during festival demand spikes.
- Closely monitor low-reliability suppliers.
- Retrain as new daily observations accumulate.

10. Conclusion

The completed workflow demonstrates multi-table data assembly, explicit data-quality treatment, operational feature engineering, leakage-aware temporal evaluation and risk-focused classification. The Random Forest substantially improves over the majority baseline while retaining strong performance on the costly Imminent class.